

CS 215: Task 01 1: Due: 08-Aug Time: as mentioned in Piazza. No extensions.

- Your answers should be in your handwriting. Try to be neat and clear. If an LLM can read the data correctly, we might be able to.
- Submission Guidelines. See **piazza** for standard static instructions.
- Your submission folder after extraction should look something like:

```
23B0019_task01/  
├─ readme.txt  
└─ task01.pdf
```

You should probably name your zip file correctly. Notice how, AFTER extraction, the two files are inside the folder containing the roll number.

0. Please write the honor code in your handwriting, with the understanding that it is true.
1. A red die, a blue die, and a yellow die (all fair six-sided dice) are rolled. Let B , R , and Y denote the numbers on the blue, red and yellow dice respectively.
- We are interested in the event $B < Y < R$.
- (a) What is the probability that no two dice land on the same number?

- (b) Given that no two dice land on the same number, what is the conditional probability that $B < Y < R$?

(c) Find $P(B < Y < R)$ using part (a).

(d) If we regard the outcome as the ordered vector (B, R, Y) , how many outcomes are there in the sample space?

(e) Without using the answer to part (c), can you determine the number of outcomes satisfying $B < Y < R$ in an alternate way?

(f) Use the results of parts (d) and (e) to verify the answer to part (c).

2. The sample mean and sample variance of five data values are, respectively,

$$\bar{x} = 104, \quad s^2 = 16.$$

If three of the data values are 102, 100, and 106, determine the other two data values.

3. A computationally efficient way to compute the sample mean and sample variance is given by

$$\bar{x}_{j+1} = \bar{x}_j + \frac{x_{j+1} - \bar{x}_j}{j+1},$$

and

$$s_{j+1}^2 = \left(1 - \frac{1}{j}\right) s_j^2 + (j+1)(\bar{x}_{j+1} - \bar{x}_j)^2.$$

- (a) Use these formulas to compute the sample mean and sample variance of

3, 4, 7, 2, 9, 6.

- (b) Verify your results in part (a) using the usual formulas for sample mean and sample variance.

- (c) Verify the recursive formula

$$\bar{x}_{j+1} = \bar{x}_j + \frac{x_{j+1} - \bar{x}_j}{j+1}.$$

4. An engineering system consisting of n independent components is said to be a k -out-of- n system if it functions whenever at least k components function.
 - (a) If the i^{th} component functions with probability P_i , $i = 1, 2, 3, 4$, compute the probability that a 2-out-of-4 system functions.
 - (b) Generalize your reasoning from part (a) to write a compact expression for the probability that a 3-out-of-5 system functions. Use summation or subset notation; do not list all outcomes individually.